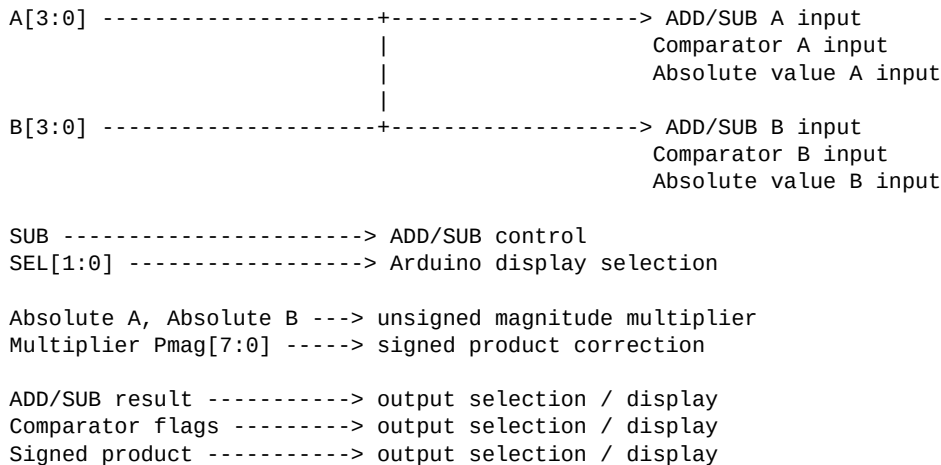


# 4-Bit Signed ALU - Architecture Explanation

This document explains the block-level architecture of the completed ALU. The design is separated into modules so each part can be tested independently.

## Top-Level Block Diagram



## ADD/SUB Block

The ADD/SUB block uses a 4-bit adder and XOR gates on the B input. The same adder performs both addition and subtraction.

```
B_eff = B XOR SUB
Result = A + B_eff + SUB
```

```
SUB = 0: Result = A + B
SUB = 1: Result = A + (~B) + 1 = A - B
```

## Absolute Value Blocks

The multiplier starts by converting A and B into magnitudes. Each magnitude block conditionally inverts the input and adds 1 if the sign bit is 1.

```
Amag = (A XOR A3) + A3
Bmag = (B XOR B3) + B3
```

## Comparator Block

The 74HC85 comparator produces unsigned outputs first. Extra logic then corrects the GT and LT outputs using the sign bits.

```
sameSign = NOT(A3 XOR B3)
GT = (NOT A3 AND B3) OR (sameSign AND U_GT)
EQ = U_EQ
LT = (A3 AND NOT B3) OR (sameSign AND U_LT)
```

## Multiplier Block

The multiplier uses the magnitude values from the absolute value blocks, performs unsigned 4x4 multiplication, then corrects the final 8-bit product sign.

```
ProductSign = A3 XOR B3
FinalProduct = (Pmag XOR ProductSign) + ProductSign
```

## Arduino Readout Layer

The Arduino readout layer is connected after the hardware logic. 74HC165 shift registers collect ADD/SUB outputs, multiplier outputs, comparator flags, SUB, and display-select inputs. The Arduino shifts these bits in, formats the selected result, and displays it on the LCD.